

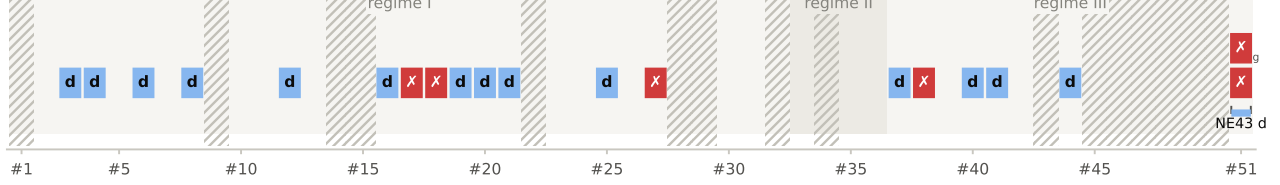
H72: Trap aging is input starvation

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 60% Faithfulness 1/5 Usefulness 2/5

Bottom line. Trap aging is intrinsic: the age clock survives every input-starvation control, and recent undirected input holds idle agents idle rather than freeing them.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



Goal periods tested (6 failed, 15 descriptive). ✔ supported ✖ failed ⚡ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

The claim: an idle agent's escape slows with trap age only because it has not read new input for a long time. Each call after an idle call is a gate, where the agent acts or idles again. Two clocks run at every gate: trap age a since the last sustained run, and starvation s since the last call that read a novel chat item (DQ1 ledger). A logit hazard with agent fixed effects holds both clocks; 18 goal periods with ≥ 200 gates; natives NE43, blocked spells and the #focus room. **The claim fails.** In all four periods that age with power, the age slope survives the control (G51: $-0.48 \rightarrow -0.53$). The starvation slope has the opposite sign: recent input lowers escape (G51 $+0.23$ [$+0.11, +0.33$]). The holdout is not run.

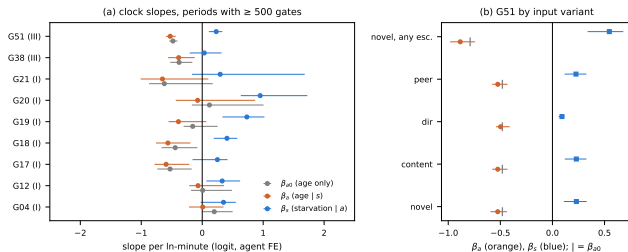
PHYSICAL MODEL

Discrete-time escape hazard on the call clock (model 09: exogenous input events; model 02: kinetic escape from the idle state). The state is idle or active per call. Current reads at the gate (directed, undirected, nudge, bookend) are kicks. Nuisance terms: declared pause, time of day, others' activity, last run length. Rivals: intrinsic aging (R1), two clocks (R2), kicks only (R3), a common room field (R4).

$$\text{logit } P(\text{esc}_g) = \alpha_i + \beta_a \ln a_g + \beta_s \ln s_g + \gamma \cdot c_g + \delta \cdot z_g$$

$$\text{H72: } \beta_a = 0, \beta_s < 0; \quad \rho = 1 - \beta_a / \beta_{a0}$$

$$b_{\text{impl}} = \beta_s \kappa, \quad \kappa = \partial \ln s / \partial \ln a \text{ (within agent)}$$



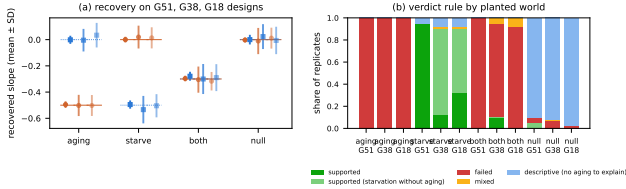
Observables. (a) Age slope without s (grey), with s (orange) and the starvation slope (blue), periods with ≥ 500 gates, day-block 95% CIs. Orange sits on grey: starvation absorbs no aging. Blue is ≥ 0 wherever it is resolved. (b) G51 under four input definitions and the glance outcome: the age slope never moves.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not treat an aged idle agent as starved of input. Feeding it room chatter will not wake it: undirected input read in the last minutes lowers its escape odds by about 20% per e -fold of recency in #51. Kicks act only at the call that reads them. Trap depth is a property of the trap, about $a^{-0.5}$ in the odds, and it does not reset when input arrives. An operator should address the agent directly at a gate (H16, H60) and not count on ambient traffic. A scaffold builder who wants fewer deep traps should cap the trap, for example with a forced re-plan after k re-pauses, and should not add more messages. Blocked (stuck-on-task) spells behave the same way: they age by themselves whatever the input.

RESULTS VS PREDICTIONS

Prediction	Observed	
P1 aging in G51	$\beta_{a0} = -0.48 [-0.54, -0.41]$	✓
P2 aging survives s	$\beta_a = -0.53, \rho = -0.09$; but $\beta_s = +0.23$	~
P3 directed s absorbs part	$\beta_{s,dir} = +0.09, \rho = -0.04$	×
P4 supported $\leq \frac{1}{3}$	0/4 powered aging periods	✓
P5 regime I unresolved	10/12 descriptive	✓
P6 CV: age > starvation	14.9 vs 1.3 nats/1000 gates	✓
P7 in-flight placebo ≈ 0	16/18; G51 +0.16 (A2, post hoc)	~
N1 NE43	not a starvation step ($s_{dir} \times 0.97$)	~
N2 blocked spells	G51 -0.54, G27 -0.70, s adds 0	✓



Second figure. Parametric bootstrap on the real G51, G38 and G18 gate designs. (a) Planted age and starvation slopes are recovered within 0.02. (b) The verdict rule calls intrinsic aging “failed” in every replicate and a null world “supported” in $\leq 5\%$. A pure starvation world yields almost no marginal aging (-0.06 to -0.10).

GOAL PERIODS IN DETAIL

G03 d descriptive
G04 d descriptive
G06 d descriptive
G08 d descriptive
G12 d descriptive
G16 d descriptive
G17 × failed
G18 × failed
G19 d descriptive
G20 d descriptive
+11 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The slopes are conditional associations at the gate; an unobserved agent state that makes an agent both read less and wake more could produce $\beta_s > 0$. Trap age is measured from the last sustained run. Fourteen of 18 periods have too little aging to test. The planned split of β_s by model family was not run. The first in-flight placebo window was outcome-dependent (pause timers); A2 fixed it after the first results, and no clock or verdict moved.

ROUND 2

Model the gate as a race between the agent’s timer and the undirected-chatter rate, with mentions as the override. Test the context self-share (H16-R1’s urn) as the aging variable on the same gates. Run `confirm.py` on the #51 tail, #45, #47–#50 and #32.

- **H72-R1** Chatter hold as the object.
- **H72-R2** What does age measure?
- **H72-R3** Run analysis/confirm.py on the #51 tail, #45, #47–#50 and #32 after commit and LOG disclosure.